

第七章 狭义相对论

§ 7.1 基本原理

1. Galileo 相对性原理

Newton 规律: $\vec{F} = m \frac{d^2 \vec{x}}{dt^2}$

Galileo 变换: $\begin{cases} \vec{x}' = \vec{x} - \vec{v}t \\ t' = t \end{cases}$

速度叠加原理: $\vec{v}' = \frac{d\vec{x}'}{dt'} = \frac{d\vec{x}}{dt} - \vec{v} \frac{dt}{dt'} = \vec{v} - \vec{v}$

若: ① 质量不依赖速度, $m = m_0$

② 作用力不依赖参考系 $\vec{F}' = \vec{F}$

则 Newton 规律在 Galileo 规律下不变.

2. 局限性

$$\vec{v} \times \vec{F} = \frac{\partial H}{\partial \vec{v}} \Rightarrow \frac{\partial H}{\partial x} - \frac{\partial H}{\partial y} = \frac{\partial D}{\partial t}$$

$\vec{z} \rightarrow \vec{z}': \vec{z}' = \vec{z} - \vec{v}t, t' = t$ 沿 $\vec{z}$ 方向做 Galileo 变换

$$\frac{\partial H}{\partial x} - \frac{\partial H}{\partial y} = \frac{\partial H}{\partial x} - \frac{\partial H}{\partial y}$$

$$\frac{\partial}{\partial t} (D(\vec{z}, t)) = \frac{\partial}{\partial t} D(\vec{z}' + \vec{v}t', t')$$

$$= \frac{\partial D}{\partial \vec{z}'} \frac{\partial \vec{z}'}{\partial t} + \frac{\partial D}{\partial t'} \frac{\partial t'}{\partial t} = \frac{\partial D}{\partial t'} - v \frac{\partial D}{\partial \vec{z}'} \neq \frac{\partial D}{\partial t'}$$

Maxwell 方程组在 Galileo 变换后不成立.

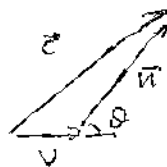
3. Michelson-Morley 实验.

假定存在以太: 在以太中光速为 c , 地球相对以太速度为 v , 则地球光速为 u

$$c^2 = (u + v \cos \theta)^2 + (v \sin \theta)^2$$

$$u = \sqrt{c^2 - v^2 \sin^2 \theta} - v \cos \theta$$

$$u = c - v \quad (\theta = 0), \quad c + v \quad (\theta = \pi), \quad \sqrt{c^2 - v^2} \quad (\theta = \frac{\pi}{2})$$



$$t_1 = l_1 \left(\frac{1}{c+v} + \frac{1}{c-v} \right) = \frac{2l_1}{c} \frac{1}{1-\beta^2} \quad \beta = \frac{v}{c}$$

$$t_2 = \frac{2l_2}{\frac{c^2-v^2}{c}} = \frac{2l_2}{c} \frac{1}{1-\beta^2}$$

$$\Delta t = t_1 - t_2 = \frac{2}{c} \left[l_1(1+\beta^2) - l_2(1+\frac{1}{\beta^2}\beta^2) \right]$$

将仪器旋转 90°

$$\text{于洛伦兹: } \Delta L = \frac{c}{\lambda_0} (\Delta t' - \Delta t) = -\frac{1}{\lambda_0} (l_1 + l_2) \beta^2$$

4. Einstein 相对性原理.

1. 一切物理定律在不同的惯性系中有相同的形式.

2. 光速在任何惯性参考系下都相同

3. Lorentz 变换

$t_0 = 0, t'_0 = 0$, 此时原点对齐

设在 t 时刻, S, S' 系中发生了事件 P , 其

时空坐标为 $(x_1, y_1, z_1, t_1), (x_2, y_2, z_2, t_2)$

(x'_1, y'_1, z'_1, t'_1) 和 (x'_2, y'_2, z'_2, t'_2)

定义事件间隔为

$$\begin{cases} \Delta S = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2 - c^2(t_2 - t_1)^2 \\ \Delta S' = (x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2 - c^2(t'_2 - t'_1)^2 \end{cases}$$

$\Delta S > 0$, 类空间隔, $\Delta S < 0$, 类时间隔

设 $t_1 = 0$ 时在 S 系中发生事件 P_1 , t 时刻在 S' 中发生事件 P_2

$P_1: (0, 0, 0, 0), P_2: (x', y', z', t')$

$$\text{间隔 } \begin{cases} S^2 = x^2 + y^2 + z^2 - c^2 t^2 \\ S'^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2 \end{cases}$$

$$S'^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2$$

S' 沿 x/x' 轴运动: $y = y', z = z'$

在 S 系中, O' 点: $x = vt$ ($x' = 0$); 在 S' 系中, O 点: $x' = -vt'$ ($x = 0$)

$S \leftrightarrow S'$ 变换是线性的.

$$\begin{cases} x = \alpha(x' + vt') \\ x' = \alpha(x - vt) \end{cases} \quad \text{所谓“解”} \quad x' = \alpha(x - vt)$$

$$x' = \alpha(x - vt) \quad t' = \frac{1}{c} \left[(1 - \alpha v) x + \alpha c^2 t \right]$$

$$x' = \frac{1}{\delta} \delta \delta' (x - vt)$$

$$S^2 = S'^2 \rightarrow x^2 - c^2 t^2 = x'^2 - c^2 t'^2$$

$$x^2 - c^2 t^2 = \delta^2 (x - vt)^2 - \frac{c^2}{\delta^2 v^2} \left[(1 - \delta \delta') x + \delta \delta' vt \right]^2$$

$$= \delta^2 x^2 - 2\delta^2 xvt + \delta^2 v^2 t^2 - \left\{ \frac{(1 - \delta \delta')^2}{\delta^2 v^2} c^2 x^2 + 2 \frac{\delta(1 - \delta \delta')}{\delta' v} c x t + \delta^2 c^2 t^2 \right\}$$

$$= \left[\delta^2 - \frac{(1 - \delta \delta')^2}{\delta'^2 v^2} \right] x^2 - \delta^2 c^2 (1 - \beta^2) t^2 - \left[2\delta^2 v + 2\delta \frac{(1 - \delta \delta')}{\delta' v} c \right] xt$$

$$= \delta^2 \left[1 - \frac{1}{\beta^2} \left(1 - \frac{1}{\delta \delta'} \right)^2 \right] x^2 - \delta^2 c^2 (1 - \beta^2) t^2 - 2\delta^2 v \left[1 + \frac{1}{\beta^2} \left(1 - \frac{1}{\delta \delta'} \right) \right] xt$$

Pr 17: 光信号发射/接收 $\rightarrow S^2 = S'^2 = 0$

$$\begin{cases} \delta^2 \left[1 - \frac{1}{\beta^2} \left(1 - \frac{1}{\delta \delta'} \right)^2 \right] - 1 = 0 \\ 1 - \delta^2 (1 - \beta^2) = 0 \\ 2\delta^2 v \left[1 + \frac{1}{\beta^2} \left(1 - \frac{1}{\delta \delta'} \right) \right] = 0 \end{cases}$$

$$\Rightarrow \delta = \frac{1}{\sqrt{1 - \beta^2}}, \quad \left[1 + \frac{1}{\beta^2} \left(1 - \frac{1}{\delta \delta'} \right) \right] = 0$$

$$\text{解得 } \delta' = \delta = \frac{1}{\sqrt{1 - \beta^2}}$$

$$\text{Lorentz 变换 } \begin{cases} x' = \delta(x - vt) \\ y' = y, z' = z \\ t' = \delta \left(t - \frac{v}{c^2} x \right) \end{cases}$$

事件间隔为 Lorentz 不变量

$$S^2 = x^2 + y^2 + z^2 - c^2 t^2$$

$$S'^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2$$

$$= y^2 + z^2 + \delta^2 (x^2 - 2xvt + v^2 t^2) - c^2 \delta^2 \left(t^2 - \frac{2vt}{c^2} x + \frac{v^2 x^2}{c^4} \right)$$

$$= y^2 + z^2 + \delta^2 \left(1 - \frac{v^2}{c^2} \right) x^2 + \delta^2 (v^2 - c^2) t^2$$

$$= y^2 + z^2 + \frac{-\beta^2}{1-\beta^2} x^2 - \frac{-\beta^2}{1-\beta^2} c^2 t^2 = S^2$$

证明：对于类空/类时间隔的两个事件，总可以找到一个惯性系，使他们同时/同地发生。

$$\text{证： } (x, y, z, t) \rightarrow (x', y', z', t')$$

$$S^2 = x^2 + y^2 + z^2 - c^2 t^2 \quad \begin{cases} > 0 \text{ 类空} \\ < 0 \text{ 类时} \end{cases}$$

$$\text{类空 - 同时, } \underline{t' = 0} \rightarrow x'^2 + y'^2 + z'^2 = S'^2 = S^2 > 0$$

$$\text{类时 - 同地, } \underline{x' = y' = z' = 0} \rightarrow -c^2 t'^2 = S'^2 = S^2 < 0$$

总存在这样的变换 (1)

6. 速度变换公式

$$\begin{cases} dx' = \gamma(dx - v dt) \\ dt' = \gamma(dt - \frac{v}{c^2} dx) \end{cases}$$

$$dx' = \gamma(dx - v dt)$$

$$u_x' = \frac{dx'}{dt'} = \frac{dx - v dt}{dt - \frac{v}{c^2} dx} = \frac{u_x - v}{1 - \frac{vu_x}{c^2}}$$

$$u_y' = \frac{dy'}{dt'} = \frac{u_y}{\gamma(1 - \frac{vu_x}{c^2})}$$

$$u_z' = \frac{dz'}{dt'} = \frac{u_z}{\gamma(1 - \frac{vu_x}{c^2})}$$

例：求解介质中的光速和折射率

Σ : 实验室系

Σ' : 介质静止系，运动速度为 v

在 Σ' 系中，光沿各方向的速度均为 c/n

在 Σ 系中：

$$u_x = \frac{u_x' + v}{1 + \frac{vu_x'}{c^2}} = \frac{1 + \beta n}{n + \beta} c$$

$$u_y = \frac{1}{\gamma} \frac{u_y'}{1 + \frac{vu_x'}{c^2}} = \frac{1}{\gamma} \frac{1}{n + \beta} c$$

$$u_{-x} = \frac{u_x' + v}{1 + \frac{vu_x'}{c^2}} = \frac{-1 + \beta n}{n - \beta} c$$

n 在各方向不同。

§ 7.2 相对论的时空性质

1. 同时的相对性.

在 S' 系中同时发生的事件, 在 S 系中不同时发生.

设事件 P_1, P_2 在 S, S' 系中的坐标为

$$(x_1, y_1, z_1, t_1), (x'_1, y'_1, z'_1, t'_1)$$

$$(x_2, y_2, z_2, t_2), (x'_2, y'_2, z'_2, t'_2)$$

$$\Delta t = t_2 - t_1$$

$$= \gamma(t'_2 + \frac{vx'_2}{c^2}) - \gamma(t'_1 + \frac{vx'_1}{c^2})$$

$$= \gamma \Delta t' + \gamma \frac{v \Delta x'}{c^2} = \frac{\gamma v \Delta x'}{c^2}$$

2. 钟慢现象.

在 S' 系中, 一静止物体内部发生两事件, 时刻为 t'_1, t'_2

$$\text{固有周期: } \Delta t = t'_2 - t'_1$$

$$\Delta s'^2 = -c^2 \Delta t'^2$$

$$= \Delta s^2 = \Delta x^2 - c^2 \Delta t^2 \quad \Delta x = v \Delta t$$

$$= v^2 \Delta t^2 - c^2 \Delta t^2$$

$$\text{解得 } \Delta t = \gamma \Delta t'$$

3. 钟慢现象的相对性.

在 S 中, C_1, C_2 校准, $\Delta t = \Delta t'$

在 S' 看来: ~~不是~~ $\Delta t = \gamma \Delta t'$

因为 C_1, C_2 并未校准

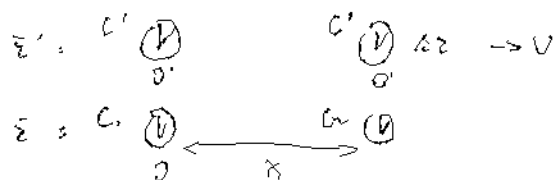
设 O' 与 O 重合校准 C_1, C_2 时, C_2 的读数 δ

在 S' 系中, C_1, C_2 同时: $0 = \gamma(\delta - \frac{vx}{c^2}) \rightarrow \delta = \frac{vx}{c^2}$

当 O' 到达 C_2 时, 在 S' 系看 S 系时钟 C_2 所经过时间

$$\Delta t' = \Delta t - \delta = \gamma \Delta t - \frac{vx}{c^2} = (\gamma - \beta^2) \Delta t = \frac{\Delta t}{\gamma}$$

故有 $\Delta t = \gamma \Delta t'$



4. 尺缩现象

由于同时的相对性, 在不同参考系中测得长度不同.

S' : 物体静止系, 固有长度为 l_0

S : 实验室系

定义 S 中测量物体的长度:

物体的前、后端分别同时经过点 P_1, P_2 , 则 $l = x_{P_2} - x_{P_1}$

Lorentz 变换:

$$x_{P_1}' = \gamma(x_{P_1} - vt_1), \quad x_{P_2}' = \gamma(x_{P_2} - vt_2)$$

$$t_1 = t_2 \quad (P_1, P_2 \text{ 同时})$$

$$l = x_{P_2} - x_{P_1} = \frac{1}{\gamma}(x_{P_2}' - x_{P_1}') = \frac{l_0}{\gamma}$$

* 尺缩现象的相对性.

5. 因果律对速度的限制.

因: $P_1(x_1, y_1, z_1, t_1)$

果: $P_2(x_2, y_2, z_2, t_2)$

在 S' 系中:

$$t_2' - t_1' = \gamma(t_2 - t_1) - \frac{\gamma v}{c^2}(x_2 - x_1)$$

设在 S 系中, 因果间信号传递速度为 u

$$x_2 - x_1 = u(t_2 - t_1)$$

$$t_2' - t_1' = \gamma(1 - \frac{uv}{c^2})(t_2 - t_1) \geq 0$$

$$\Rightarrow uv \leq c^2, \text{ 当 } v < c, \text{ 则 } u \leq c$$

一切信号传递速度都不能超光速

6. 四维时空

$$\text{引入 } x_4 = ict$$

$$\text{定义 } |\vec{x}|^2 = |(x_1, x_2, x_3, x_4)|^2 = S^2$$

$$= |\vec{x}'|^2 = S'^2 \quad \vec{x} \xrightarrow{L} \vec{x}'$$

$$\text{Minkowski 坐标 } (x_\mu) = (x_1, x_2, x_3, x_4)$$

$$S^2 = x_\mu x_\mu$$

$$= \frac{-\delta V}{\delta t}$$

Lorentz 变换

$$\begin{pmatrix} x_1' \\ x_2' \\ x_3' \\ x_4' \end{pmatrix} = \begin{pmatrix} \sigma & 0 & 0 & i\delta\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i\delta\beta & 0 & 0 & \sigma \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

$$x_{\mu'} = L_{\mu\nu} x_\nu$$

正交性: $L_{\mu\nu} L_{\mu\alpha}$

$$= \begin{pmatrix} \sigma & 0 & 0 & i\delta\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i\delta\beta & 0 & 0 & \sigma \end{pmatrix} \begin{pmatrix} \sigma & 0 & 0 & i\delta\beta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i\delta\beta & 0 & 0 & \sigma \end{pmatrix}$$

$$= \text{diag} (\sigma^2 + [i\delta\beta]^2, 1, 1, [-i\delta\beta]^2 + \sigma^2)$$

$$= \text{diag} (1, 1, 1, 1) = \delta_{\mu\alpha}$$

§ 7.3 电动力学规律的协变形式

1. Lorentz 张量

零阶：Lorentz 标量 $c, q, s^2, \dots$

一阶：Lorentz 矢量 $U_\mu = L_{\mu\nu} U_\nu$

二阶：Lorentz 张量 $T_{\mu\nu} = L_{\mu\alpha} L_{\nu\beta} T_{\alpha\beta}$

例：证明两个四维矢量的点积是 Lorentz 标量。

$$a = A_\mu B_\nu \delta_{\mu\nu}$$

$$a' = A'_\alpha B'_\beta \delta_{\alpha\beta}$$

$$= L_{\alpha\mu} A_\mu L_{\beta\nu} B_\nu \delta_{\alpha\beta}$$

$$= L_{\alpha\mu} L_{\alpha\nu} A_\mu B_\nu = A_\mu B_\nu \delta_{\mu\nu}$$

标积在 Lorentz 变换下不变。

一些 Lorentz 不变量：

d'Alembert 算符： $\square^2 = (\nabla^2, -c^2 \frac{\partial^2}{\partial t^2}) = \partial_\mu \partial_\mu$

事件间隔： $ds^2 = dx_\mu dx_\mu = (dx^2, -cdt^2)$

时空体积元： $dx'_1 dx'_2 dx'_3 dx'_4 = \left| \frac{\partial x'}{\partial x} \right| dx_1 dx_2 dx_3 dx_4$

$$= |\det L| dx_1 dx_2 dx_3 dx_4$$

$$= dx_1 dx_2 dx_3 dx_4$$

2. 电荷守恒定律的协变形式

定义 $(J_\mu) = (J_1, J_2, J_3, ic\rho)$

$$\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$$

$$\frac{\partial J_1}{\partial x_1} + \frac{\partial J_2}{\partial x_2} + \frac{\partial J_3}{\partial x_3} + \frac{\partial (ic\rho)}{\partial (ict)} = 0$$

$$\partial_\mu J_\mu = 0 \quad \partial_\mu = \frac{\partial}{\partial x_\mu}$$

Lorentz 标量。

3. d'Alembert 方程的协变形式、四维势

在 Lorentz 规范下:

$$\begin{cases} \square^2 \vec{A} = -\mu_0 \vec{j} \\ \square^2 \varphi = -\mu_0 c^2 \rho \end{cases}$$

在 4 维势 $(A_\mu) = (A_1, A_2, A_3, \frac{i\varphi}{c})$

$$\square^2 A_\mu = -\mu_0 j_\mu$$

Lorentz 规范: $\partial_\mu A_\mu = 0$

例: 实验室参考系里有一匀速运动的点电荷 q , 求其产生的四维势

在 S' 系中: $(A'_\mu) = (0, 0, 0, \frac{iq}{4\pi\epsilon_0 c r'})$

$$A_\mu = L_{\mu\nu}^{-1} A'_\nu = L_{\mu\nu}^T A'_\nu$$

$$= \begin{pmatrix} 0 & 0 & 0 & -i\beta\mu \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ i\beta\mu & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{iq}{4\pi\epsilon_0 c r'} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{i\beta q}{4\pi\epsilon_0 c r'} \\ 0 \\ 0 \\ \frac{i\beta q}{4\pi\epsilon_0 c r'} \end{pmatrix}$$

$$r' = [x'^2 + y'^2 + z'^2]^{1/2}$$

$$= [\gamma^2 (x - vt)^2 + y^2 + z^2]^{1/2}$$

4. Doppler 效应和光行差, 波的相位是 Lorentz 不变量.

相位: $\phi = \vec{k} \cdot \vec{x} = \vec{k} \cdot \vec{x} - \omega t$

$$= \vec{k} \cdot \vec{x} + (i\frac{\omega}{c})(ict)$$

$$(k_\mu) = (k_1, k_2, k_3, \frac{i\omega}{c})$$

$$\phi = k_\mu x_\mu$$

$$k'_\mu = L_{\mu\nu} k_\nu$$

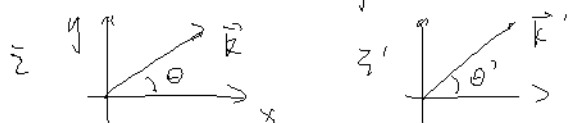
$$k'_1 = \gamma(k_1 - v k_4) = \gamma k_1$$

$$k'_4 = \gamma(k_4 - v k_1) = \gamma \frac{i\omega}{c}$$

$$\begin{cases} k_x' = k_x + \gamma\beta \frac{\omega}{c} \\ k_y' = k_y, \quad k_z' = k_z \\ \frac{i\omega'}{c} = -i\beta \gamma k_x + \gamma \cdot \frac{i\omega}{c} \end{cases} \rightarrow \begin{cases} k_y' = k_y, \quad k_z' = k_z \\ \omega = \gamma(1 - v k_x + \omega) \end{cases}$$

$$\begin{cases} \vec{k}_\parallel' = \gamma(\vec{k}_\parallel - \frac{\omega}{c} \vec{v}) \\ \vec{k}_\perp' = \vec{k}_\perp \\ \omega' = \gamma(-\vec{v} \cdot \vec{k} + \omega) \end{cases} \quad \text{逆变换:} \quad \begin{cases} \vec{k}_\parallel = \gamma(\vec{k}_\parallel' + \frac{\omega'}{c} \vec{v}) \\ \vec{k}_\perp = \vec{k}_\perp' \\ \omega = \gamma(\vec{v} \cdot \vec{k}' + \omega') \end{cases}$$

假设在参考系 Σ 中, 光源静止.



$$\begin{cases} k_x = k_0 \cos \theta = \frac{\omega}{c} \cos \theta \\ k_x' = k_0' \cos \theta' = \frac{\omega'}{c} \cos \theta' \end{cases} \quad \begin{cases} k_y = k_0 \sin \theta = \frac{\omega}{c} \sin \theta \\ k_y' = k_0' \sin \theta' = \frac{\omega'}{c} \sin \theta' \end{cases}$$

$$\begin{cases} \omega' \cos \theta' = \gamma(\omega \cos \theta - \beta \omega) \\ \omega' \sin \theta' = \omega \sin \theta \end{cases}$$

角得 $\tan \theta' = \frac{\sin \theta}{\gamma(\cos \theta - \beta)}$, $\omega' = \omega \sqrt{\gamma^2(\cos \theta - \beta)^2 + \sin^2 \theta} = \omega \gamma(1 - \beta \cos \theta)$

① $\theta = 0$ $\omega = \omega_0 \frac{1}{\gamma(1-\beta)} = \omega_0 \sqrt{\frac{1+\beta}{1-\beta}}$, $\Delta \omega \approx \beta \omega_0$

② $\theta = 90^\circ$ $\omega = \frac{1}{\gamma} \omega_0$ $\Delta \omega \approx -\frac{1}{2} \beta^2 \omega_0$ 较弱.

③ $\theta = 180^\circ$ $\omega = \omega_0 \sqrt{\frac{1-\beta}{1+\beta}}$, $\Delta \omega \approx -\beta \omega_0$

④ θ 任意 $\omega = \omega_0 \frac{1}{\gamma(1-\beta \cos \theta)}$, $\Delta \omega \approx \beta \cos \theta \omega_0$

* 验证时间膨胀:

$$\omega_1 = \frac{\omega_0}{\gamma(1-\beta)}, \quad \omega_2 = \frac{\omega_0}{\gamma(1+\beta)}, \quad \omega_1 \omega_2 = \omega_0^2$$

Galileo 原理给出:

$$\omega_1' = \omega|_{\theta=0} = \frac{\omega_0}{1-\beta}, \quad \omega_2' = \omega|_{\theta=180^\circ} = \frac{\omega_0}{1+\beta},$$

$$\omega_1' \omega_2' = \omega_0^2 \frac{1}{1-\beta^2} \approx \omega_0^2 (1+\beta^2)$$

验证 $\omega_1 \omega_2 / \omega_0^2 \approx 1$

5. Doppler效应在用，测定液体流速。

粒子感受到的入射光频率

$$\omega = \omega_0 \gamma \left[1 - \beta \cos\left(\frac{\pi}{2} + \theta\right) \right]$$

$$= \gamma (1 \pm \beta \sin \theta) \omega_0$$



散射光频率：

$$\omega' \approx \omega = \gamma (1 \pm \beta \sin \theta) \quad \text{忽略横向 Doppler 效应, } O(\beta^2)$$

干涉频率， $\Delta\omega = \omega'_+ - \omega'_- = 2\gamma\beta \sin \theta \omega_0$

$$f = \frac{\Delta\omega}{2\pi} = \frac{\gamma\beta \sin \theta}{\pi} \omega_0$$

6. 物理定律的协变形式

比如 $G_{\mu\nu} = G_{\mu\alpha} A_\alpha + B_{\mu\nu}$ 的方程

$$G'_\mu = L_{\mu\nu} G_\nu$$

$$= L_{\mu\nu} (G_{\nu\alpha} A_\alpha + B_{\nu\mu}) = L_{\mu\nu} G_{\nu\alpha} \delta_{\lambda\beta} A_\beta + L_{\mu\nu} B_{\nu\mu}$$

$$= L_{\mu\nu} G_{\nu\alpha} L_{\lambda\alpha} L_{\lambda\beta} A_\beta + L_{\mu\nu} B_{\nu\mu}$$

$$= (L_{\mu\nu} L_{\lambda\alpha} G_{\nu\alpha}) (L_{\lambda\beta} A_\beta) + L_{\mu\nu} B_{\nu\mu} = G'_{\mu\lambda} A'_\lambda + B'_{\mu\nu}$$

7. 电磁场张量

Maxwell 方程也具有协变性。

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial t} = -\nabla \left(\frac{c\varphi}{c} \right) - ic \frac{\partial \vec{A}}{\partial (ict)}$$

$$= ic \left[\nabla \left(\frac{c\varphi}{c} \right) - \frac{\partial \vec{A}}{\partial (ict)} \right]$$

$$\left\{ \begin{array}{l} B_1 = \partial_2 A_3 - \partial_3 A_2 \\ B_2 = \partial_3 A_1 - \partial_1 A_3 \end{array} \right.$$

$$\left\{ \begin{array}{l} E_1 = ic(\partial_2 A_4 - \partial_4 A_2) \\ E_2 = ic(\partial_3 A_4 - \partial_4 A_3) \end{array} \right.$$

$$B_3 = \partial_1 A_2 - \partial_2 A_1$$

$$E_3 = i c (\partial_2 A_4 - \partial_4 A_2)$$

$(\vec{E}, \vec{B})$ 表示为四矢量 $(\vec{A}, \frac{i\varphi}{c})$ 的广义旋度.

定义: $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ 二阶反对称张量.

$$= \begin{pmatrix} 0 & B_3 & -B_2 & -\frac{iE_1}{c} \\ -B_3 & 0 & B_1 & -\frac{iE_2}{c} \\ B_2 & -B_1 & 0 & -\frac{iE_3}{c} \\ \frac{iE_1}{c} & \frac{iE_2}{c} & \frac{iE_3}{c} & 0 \end{pmatrix}$$

$$F_{\mu\nu} F^{\mu\nu} = F_{\mu\nu} F_{\nu\mu}^T = (F F^T)_{\mu\mu} = \text{Tr}(F F^T) = 2(\vec{B}^2 - \frac{\vec{E}^2}{c^2})$$

$$\det(F) = \frac{1}{c^2} [B_1 B_2 E_1 E_2 - B_1^2 E_3^2 + B_1 B_2 E_1 E_2 + B_2^2 E_3^2 + B_1^2 E_3^2] \\ = -\frac{1}{c^2} (\vec{B} \cdot \vec{E})$$

$$\vec{B}^2 - \frac{1}{c^2} \vec{E}^2, \vec{B} \cdot \vec{E} \text{ 是 Lorentz 不变量}$$

如 Maxwell 方程组的协变形式.

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\frac{\partial E_3}{\partial x_1} = i c \frac{\partial F_{14}}{\partial x_3} = -i c \mu_0 J_4 \rightarrow \frac{\partial F_{10}}{\partial x_3} = \mu_0 J_4 \quad F_{44} = 0$$

$$(\nabla \times \vec{B} - \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t})_i = \mu_0 J_i \quad i = 1, 2, 3$$

$$F_{11} = 0$$

$$\frac{\partial B_3}{\partial x_1} - \frac{\partial B_2}{\partial x_3} - \frac{1}{c} \frac{\partial E_1}{\partial (ict)} = \frac{\partial F_{12}}{\partial x_3} - \frac{\partial (-F_{12})}{\partial x_3} + \frac{\partial F_{14}}{\partial x_4} = \frac{\partial F_{14}}{\partial x_4} = \mu_0 J_1$$

$$\hookrightarrow \frac{\partial F_{k0}}{\partial x_0} = J_k \quad k = 1, 2, 3$$

$$\begin{cases} \nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \\ \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{cases} \Rightarrow \frac{\partial F_{k0}}{\partial x_0} = \mu_0 J_k$$

$$\begin{cases} \vec{\nabla} \cdot \vec{B} = 0 \\ \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \end{cases} \Rightarrow \frac{\partial F_{10}}{\partial x_1} + \frac{\partial F_{01}}{\partial x_1} + \frac{\partial F_{10}}{\partial x_1} = 0$$

9. 电磁场的变换.

$$F_{\mu\nu} = L_{\mu\alpha} L_{\nu\beta} F_{\alpha\beta} = (L F L^T)_{\mu\nu}$$

$$= \begin{pmatrix} \gamma & 0 & 0 & -i\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ i\beta\gamma & 0 & 0 & \gamma \end{pmatrix} \begin{pmatrix} 0 & B_1 & -B_2 & -\frac{iE_1}{c} \\ -B_1 & 0 & B_2 & -\frac{iE_2}{c} \\ B_2 & -B_1 & 0 & -\frac{iE_3}{c} \\ \frac{iE_1}{c} & \frac{iE_2}{c} & \frac{iE_3}{c} & 0 \end{pmatrix} \begin{pmatrix} \gamma & 0 & 0 & i\beta\gamma \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -i\beta\gamma & 0 & 0 & \gamma \end{pmatrix}$$

$$= \begin{pmatrix} \frac{\partial E_1}{\partial t} & \gamma B_1 + \frac{\partial E_2}{\partial t} & -\gamma B_2 + \frac{\partial E_1}{\partial t} & -\frac{\partial E_1}{\partial t} \\ \dots & \dots & \dots & \dots \end{pmatrix}$$

$\vec{E} \rightarrow \vec{E}'$,

$$\begin{cases} E'_1 = E_1 \\ E'_2 = \gamma(E_2 - vB_3) \\ E'_3 = \gamma(E_3 + vB_2) \end{cases} \begin{cases} B'_1 = B_1 \\ B'_2 = \gamma(B_2 + \frac{v}{c^2} E_3) \\ B'_3 = \gamma(B_3 - \frac{v}{c^2} E_2) \end{cases}$$

$$\begin{cases} E'_1 = E_{11}, \quad \vec{E}'_2 = \gamma(\vec{E}_2 + \vec{v} \times \vec{B}_2) \\ B'_1 = B_{11}, \quad \vec{B}'_2 = \gamma(\vec{B}_2 - \frac{\vec{v}}{c^2} \times \vec{E}_2) \end{cases}$$

例：计算实验室中匀速运动的点电荷 q 所产生的电磁场

$$\text{在 } z' \text{ 系中: } \vec{E}' = \frac{q}{4\pi\epsilon_0} \frac{\vec{r}'}{r'^3}, \quad \vec{B}' = 0$$

$$\vec{z}' \rightarrow \vec{z}$$

$$E_x = E'_x = \frac{q}{4\pi\epsilon_0} \frac{x'}{r'^3}, \quad B_x = 0$$

$$E_y = \gamma(E'_y - v B'_z) = \frac{\gamma q}{4\pi\epsilon_0} \frac{y'}{r'^3}, \quad E_z = \frac{\gamma q}{4\pi\epsilon_0} \frac{z'}{r'^3}$$

$$B_y = \frac{\gamma v}{c^2} E'_z = \frac{\gamma q v}{4\pi\epsilon_0 c^2} \frac{z'}{r'^3}, \quad B_z = -\frac{\gamma q v}{4\pi\epsilon_0 c^2} \frac{y'}{r'^3}$$

$$x' = \gamma(x - vt), \quad y' = y, \quad z' = z$$

$$r' = \sqrt{x'^2 + y'^2 + z'^2}$$

$$\vec{E} = (E_x, E_y, E_z)$$

$$= \frac{q}{4\pi\epsilon_0} \frac{1}{[\gamma^2(x-vt)^2 + y^2 + z^2]^{3/2}} (\gamma(x-vt), \gamma y, \gamma z)$$

记 $\vec{R} = (x-vt, y, z)$ 为 z 系中, q 在 t 时刻的位置

$$= \frac{\gamma q}{4\pi\epsilon_0 R^3} \left[1 - \left(\frac{x}{R} \right)^2 \gamma^2 \beta^2 \right]^{3/2} \vec{R}$$

$$= \frac{q}{4\pi\epsilon_0 R^3} \left(1 - \beta^2 \sin^2 \theta \right)^{3/2} \vec{R}$$

例: 在 $\bar{z}'$ 系中,

$$\vec{k}_z' = \gamma (k_z - \frac{v}{c^2} \omega_z) = \gamma (1 - \beta) k_z \vec{e}_x$$

$$\omega_z' = \gamma (\omega_z - v k_z) = \gamma (1 - \beta) \omega_z$$

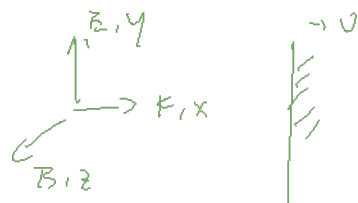
$$\vec{E}_z = E_{z0} e^{i(k_z x - \omega_z t)} \vec{e}_y$$

$$\vec{B}_z = \frac{1}{c} E_z \vec{e}_z$$

$$\vec{E}_z' = \gamma (\vec{E}_z - v \vec{B}_z) = \gamma (1 - \beta) E_z \vec{e}_x$$

$$\vec{B}_z' = \gamma (\vec{B}_z - \frac{1}{c^2} \vec{v} \times \vec{E}_z)$$

$$\vec{B}_z' = \frac{1}{c} (1 - \beta) E_z = \gamma (1 - \beta) B_z \vec{e}_z$$



反射后:

$$\vec{k}_z' = \vec{k}_z', \quad \omega_z' = \omega_z', \quad \vec{E}_z = -\vec{E}_z', \quad \vec{B}_z = -\vec{B}_z',$$

$$\vec{E}_z = -E_z' e^{i(-k_z' x - \omega_z' t)} \vec{e}_y$$

$$\vec{B}_z = -\frac{1}{c} E_z' \vec{e}_z$$

$\bar{z}' \rightarrow \bar{z}$:

$$-k_z = \gamma (-k_z' + \frac{v}{c^2} \omega_z')$$

$$\begin{aligned} \Rightarrow k_z &= \gamma (k_z' - \frac{v}{c^2} \omega_z') = \gamma k_z' (1 - \beta) \\ &= \gamma^2 (1 - \beta)^2 k_z = \frac{1 - \beta}{1 + \beta} k_z \end{aligned}$$

$$\omega_z = \gamma [\omega_z' + v (-k_z')]$$

$$= \gamma \omega_z' (1 - \beta) = \gamma^2 (1 - \beta)^2 \omega_z$$

$$= \frac{1 - \beta}{1 + \beta} \omega_z$$

$$\vec{E}_z = \gamma (\vec{E}_z' + \vec{v} \times \vec{B}_z')$$

$$= \gamma [-E_z' \vec{e}_z + v B_z' (-\vec{e}_x) \times (-\vec{e}_z)]$$

$$= -\sigma (E'_x - v B'_y) \vec{e}_y$$

$$= -\sigma (1-\beta) E'_x = -\sigma^2 (1-\beta)^2 E_z \vec{e}_y$$

$$= -\frac{1-\beta}{1+\beta} E_z \vec{e}_y$$

$$\vec{B}_y = -\frac{c}{c} E_0 \vec{e}_z$$